Roll No. ....

TCS–509

B. TECH. (CSE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Oct., 2022

MACHINE LEARNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is Machine Learning ? Explain supervised and unsupervised learning algorithms. Give name of different algorithms used for supervised and unsupervised learning. (CO1)

OR

(b) Find out the mean, median, mode, variance and standard deviation for the height of animals : 550 mm, 450 mm, 160 mm, 410 mm, 300 mm, 450 mm and 240 mm. (CO3)

2. (a) Suppose you are working on a binary classification problem. You trained a model on a training dataset and get the below error matrix on the testing dataset : (CO2)

Predicted Class

Classes		Yes	No
Actual Class	Yes	90	210
	No	140	9560

P. T. O.

This is the confusion matrix for classes Cancer = Yes and Cancer = No. Find out the following performance metric :

(i) Sensitivity/Recall

(ii) Specificity

(iii) Precision

(iv) Accuracy

(v) False positive rate

(vi) F-score

OR

(b) Define the eigen value and eigen vector of the matrix. Find the eigen values and eigen vectors of the following matrix : (CO3)

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

3. (a) Given two objects represented by the tuples (45, 10, 48, 19) and (20, 12, 52, 18) : (CO1)

(i) Compute the Euclidean distance and Manhattan distance between the two objects.

(ii) Compute the Minkowski distance between the two objects, using $q = 3$.

(b) Which Machine Learning algorithm is known as lazy learner ? Define K-nearest neighbor algorithm with example. Also explain how k should be selected. (CO2)

4. (a) Explain the concept of entropy in decision tree learning. Discuss the necessary measure required to select the attributes for building a decision tree using ID3 algorithm. (CO1)

OR

(b) Write an algorithm for Bayesian classification. Illustrate the algorithm with a relevant example. (CO2)

5. (a) What is the purpose of training, validation and testing processes ? In real-world data, tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem. (CO1)

OR

(b) What is an attribute ? Describe the following kind of attributes : (CO3)

(i) Nominal attributes

(ii) Binary attributes

(iii) Ordinal attributes

(iv) Numeric attributes

(v) Discrete *vs*. Continuous attributes

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